

Case study



Product Manager Service Cockpit, New Business & Innovation

You are asked to build a small working version of the product you would own, using the data it would run on, and to decide for yourself what belongs in it.

Time budget

Four hours in total. This is a cap, not a target. Tell us at the end how long you actually spent.

You receive

This brief and two CSV files: `installation_base.csv` and `telemetry_daily.csv`

Tools

Any AI tooling you like, and anything else you like. Building with AI is the job, not a shortcut around it.

You send back

A prototype that runs, plus written answers of no more than three pages. Details on the last page.

Background

thermondo installs and services heat pumps in single family homes across Germany. The installed base runs into the thousands and comes from several different manufacturers. Part of the base is connected and sends operating data back, part of it is not. How much is which, you can read off the files.

Each manufacturer exposes its data differently. One is read directly over a local interface. One is reached through a gateway that exposes a restricted service role only. One cannot be reached directly at all and is read through a third party device in the house. The fields have different names, different units, different semantics, and different levels of completeness. Nobody has ever put them side by side.

Today the service organisation works in two separate tools, neither of which covers the whole installed base. When a customer calls, the person on the phone often cannot see the unit at all. When a technician drives out, they frequently learn what is wrong only once they are standing in the cellar.

There is a first read only prototype. It is a small web application that pulls a snapshot from the reporting layer every 30 minutes. It was built without an engineering team, and there is no engineering team available for the next version either. Everything in this role gets built the way you will build it in this case study.

Read only is the current state, not the plan. The stages after a single view include remote maintenance and remote changes to unit settings across the connected base. The service organisation is not the only group that wants in. Subcontracted installation partners service the same units.

On the data in this package. The two CSV files are synthetic. They were generated to carry the same structure, the same field coverage per manufacturer and the same kinds of defects as the real extract. No customer data is included: the names and addresses in the file are invented, the phone numbers are redacted, and the reference numbers are not real ones. The manufacturers are anonymised as vendor A, B and C.

Raw material

The following is written for this case study out of themes that come up again and again in the service organisation and in the ticket system. The voices and the tickets are composed, not lifted from records, and none of them is a statement by an identifiable person. What they carry is the substance we really hear. It is not a requirements document, and it is not consistent. Nobody has cleaned it up for you.

“The data is not the problem for me. When I see a flow temperature and a return temperature and how often the compressor started, I know what is going on. The problem is that for half the units I see nothing at all, and for the other half I do not know whether what I am looking at is from this morning or from last Tuesday.”

Service technician, eleven years in the field

“I have two tools open all day. One of them covers one manufacturer properly. The other one covers a bit of everything and I do not trust the numbers in it. So I call the technician who was last on site and ask him.”

Service dispatcher

“What I want in the morning is a short list. Ten, fifteen units, not four hundred. These are the ones that will turn into a complaint this week if nobody looks at them. Then I decide who drives where. If you give me a screen with every unit on it, I will not open it.”

Team lead, service planning

“Do not hide anything from me. I want every unit and every value, and I want to filter it myself. The people who summarise things for me always summarise away exactly the thing I was looking for.”

Service technician, second interview

“The thing I actually need is to change the heating curve while I still have the customer on the line. Then half my callbacks disappear.”

Second level support

“And when I can see the unit is running hot, I want to restart it from here. Or shut it down if it is doing damage. Best case for the whole list at once instead of one by one.”

Second level support, same conversation

“The link to the current tool sits in three of my chats. Somebody sent it to me months ago and it still works, on my private phone too. Our subcontracted installers keep asking whether they can have it as well, they are on the same units.”

Regional service coordinator

“What we do not need is another manual. The colleagues know their trade. Where it gets thin is the newest product line, and we are handling that with training.”

Head of service operations

Ticket 41-xxxx, closed, “no fault found”

Customer reports the heat pump is loud and the heating is not getting warm. Technician on site, unit in normal operation, no fault memory entry, nothing done. Travel and time billed internally. Note in the ticket: “customer probably means the noise during defrost.”

Ticket 41-xxxx, third visit on the same unit

Third appointment within five weeks. First visit sensor replaced, second visit expansion valve checked, third visit circulation pump replaced. Note in the ticket: “if somebody had looked at the trend, we would have seen this in week one.”

Ticket 41-xxxx, escalation

Customer without hot water for four days before calling. Unit was connected the whole time. Note from the team lead: “we could have known this on day one. Nobody was looking.”

Ticket 41-xxxx, internal note

During a remote session the wrong unit was reset. The customer was without heating for one night. Note from the team lead: “the two reference numbers differ by one digit.”

The data package

installation_base.csv, one row per installation record

Field	Description
reference_number	Installation reference. Written by several source systems over several years.
vendor	Manufacturer of the unit.
customer_name	Name of the household.
street_address	Street and house number.
contact_phone	Phone number on file. Redacted throughout this package.
postcode_region	First two digits of the postcode.
commissioning_date	Date the unit went into operation.
connectivity	Whether the unit is expected to send data.
service_tier	Service plan the customer holds.
last_service_visit	Last recorded on site visit.

telemetry_daily.csv, one row per unit per day, 1 June to 30 July 2026

Field	Description
reference_number	Reference of the unit, as delivered by the respective source.
reading_date	Day of the reading.
vendor	Manufacturer of the unit.
outdoor_temp_c	Daily mean outdoor temperature in degrees Celsius.
flow_temp_c	Daily mean flow temperature in degrees Celsius.
return_temp_c	Daily mean return temperature in degrees Celsius.
dhw_actual_c	Measured domestic hot water temperature in degrees Celsius.
electrical_energy_kwh	Electrical energy in kWh, as delivered by the vendor. The vendors do not define this field the same way and we have no documentation from them on it.
thermal_energy_kwh	Thermal energy delivered in kWh, as delivered by the vendor.
compressor_starts	Number of compressor starts on that day.
defrost_cycles	Number of defrost cycles on that day.
error_code_raw	Fault or status code exactly as the vendor sends it. Each vendor uses its own encoding. We have no code list for any of them.
status_raw	Operating status exactly as the vendor sends it.

This is everything we know about the fields. There is no further documentation, and inside thermondo nobody can currently explain all of it either. That is part of the situation.

Part 1, decide what the cockpit is for

about 45 minutes Half a page, no more.

- What is this cockpit for, in one or two sentences, and who is the primary user?
- Which requirements do you take from the raw material above? List them in the order you would build them.
- What did you deliberately leave out, and why?

Part 2, build it

about 2 hours This is the main part.

Build a working first version of the cockpit on top of the two CSV files. It has to run and it has to show real values computed from the data, not mock ups. Web application, notebook, spreadsheet with a front end, whatever you can make work in the time. Use whichever AI tools you prefer.

We are not judging code quality or visual polish. We are judging whether it runs, whether the right things are on the screen, and whether the numbers on the screen hold up when somebody checks them.

You will not get everything in. Decide what goes in and be ready to say why.

Part 3, read the data

about 45 minutes Answer in the written part, briefly.

1. How many heat pumps are actually in this dataset? Describe what you had to do to arrive at that number, and what you did with the records you could not resolve.
2. Which units would you put on a technician's list for tomorrow morning? Name them, say what made you pick them, and say how sure you are.
3. For each of these four questions, say whether the data answers it, partly answers it, or does not answer it, and why: how efficiently is a unit running, is a unit delivering hot water, is a unit heating at all, is a fault recurring or a one off?
4. Which fault signals occur most often? Which of them would you keep off the front page of the cockpit, and what made you decide that?

Part 4, what you need from other people

about 30 minutes Half a page.

- Which field or access is missing that would change the product most? Name one, not a list.
- Who has to deliver it, and what is your read on how hard that will be?
- Write the actual message you would send in your first week to get it. A few sentences, as you would really send them.

What to send us

1. The prototype. A link, or a file we can run. If it needs a step to start, write that step down. If it is reachable at a URL, tell us who can open it.
2. Written answers to parts 1, 3 and 4. Three pages maximum, any format.
3. Four screenshots or a screen recording of up to five minutes, so we can see it working even if something breaks on our machine.
4. Which AI tools you used, and one concrete thing the AI got wrong that you had to correct.
5. How long you actually spent.

One request. Do not polish this into a consulting deliverable. We would rather see three things that are correct and one honest sentence about what you did not get to, than ten things nobody has checked.

If something in the brief is unclear, make an assumption, write it down, and carry on. Questions are welcome at any point, and asking one does not count against you.

thermondo GmbH, New Business & Innovation. Case study for the position Product Manager Service Cockpit. The data in this package is synthetic and contains no customer data. Please do not pass the package on.